

SQL实验3

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1 查找比较Sci.系拥有3个学分的课程名称。 Sci.'复制到查询的条件中)

```
SELECT title
FROM course
WHERE dept_name = 'Comp. Sci.'
AND credits = 3;
```

	title
1	International Finance
2	Computability Theory
3	Japanese

2 找出教师的最高工资。

```
SELECT MAX(salary) AS max_salary
FROM instructor;
```

	max_salary
1	124651.41

3 找出所有教师的工资最高的（可能有多名教师的工资相同）。

```
SELECT *
FROM instructor
WHERE instructor.salary = (SELECT MAX(salary) FROM instructor);
```

	ID	name	dept_name	salary
1	19368	Wieland	Pol. Sci.	124651.41

4 查找2009年秋季的每个课程段的选课大学生。

```
SELECT course_id, sec_id, COUNT(ID) AS enrollment
FROM takes
WHERE year = 2009 AND semester = 'Fall'
GROUP BY course_id, sec_id;
```

	course_id	sec_id	enrollment
1	242	1	295
2	334	1	268
3	960	1	307
4	304	1	307
5	105	1	327
6	486	1	304
7	237	2	311

5 从2009年秋季的所有课程段中，查找最多的选课人数。

```
SELECT MAX(enrollment) AS max_enrollment
FROM (
    SELECT COUNT(ID) AS enrollment
    FROM takes
    WHERE year = 2009 AND semester = 'Fall'
    GROUP BY course_id, sec_id
) AS enrollment_counts;
```

	max_enrollment
1	327

6 查找在2009年秋季拥有最多选课X的课程段。

```
SELECT course_id, sec_id, semester, COUNT(ID) AS enrollment
FROM takes
WHERE year = 2009 AND semester = 'Fall'
GROUP BY course_id, sec_id, semester
HAVING COUNT(ID) = (
    SELECT MAX(enrollment)
    FROM (
        SELECT COUNT(ID) AS enrollment
        FROM takes
        WHERE year = 2009 AND semester = 'Fall'
        GROUP BY course_id, sec_id
    ) AS enrollment_counts
);
```

	course_id	sec_id	semester	enrollment
1	105	1	Fall	327

7 找出所有至少选修一门Comp. Sci.课程的学生姓名，保证结果中没有重复的姓名；

注：共866人

```
SELECT DISTINCT s.name
FROM student s
JOIN takes t ON s.ID = t.ID
JOIN course c ON t.course_id = c.course_id
WHERE c.dept_name = 'Comp. Sci.';
```

	name
1	Chanon
2	Seta
3	Randers
4	Mathias
5	Yeoh
6	Prabhakaran
7	Griffin
8	Soricu
9	Godfrey
10	Giannoulis
11	Hirakuri
12	Harmon
13	Baba
14	Rowe
15	Lin
16	Shishkin
17	Carey
18	Umeshara
19	Gonzalez
...	...

查询已成功执行。 ASUKALEC (15.0 RTM) / ASUKALEC 薛 (53) college.big 00:00:00 866 行

8 查找所有在2009年春季之前没有选修改的任何课程的学生ID和姓名；

```
SELECT ID, name
FROM student
WHERE ID NOT IN (
    SELECT ID
    FROM takes
    WHERE year < 2009
);
```

无查询结果

9 找出每个系教师的最高工资值。可以假设每个系至少有一位教师。

```
SELECT dept_name, MAX(salary) AS max_salary
FROM instructor
GROUP BY dept_name;
```

	dept_name	max_salary
1	Accounting	71351.42
2	Astronomy	79070.08
3	Athletics	103146.87
4	Biology	77036.18
5	Comp. Sci.	115469.11
6	Cybernetics	117836.50
7	Elec. Eng.	90038.09
8	English	118143.98
9	Finance	105311.38
10	Geology	99382.59
11	Languages	90891.69
12	Marketing	119921.41
13	Mech. Eng.	107978.47
14	Physics	121141.99
15	Pol. Sci.	124651.41
16	Psychology	62579.61
17	Statistics	104563.38

10 从最近查询 (i) 中所计算出的每个系最高工资中选出最低值

```

SELECT MIN(max_salary) AS min_of_max_salaries
FROM (
    SELECT dept_name, MAX(salary) AS max_salary
    FROM instructor
    GROUP BY dept_name
) AS dept_max_salaries;

```

	min_of_max_salaries
1	62579.61